

Technical Document #237: Database Systems - Comprehensive Overview

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Sharding distributes data across multiple database servers. It improves performance by parallelizing query processing.

Database indexing improves query performance by creating data structures that allow faster data retrieval.

NoSQL databases handle unstructured and semi-structured data. Document stores, key-value stores, and graph databases are common NoSQL types.

Relational databases organize data into tables with rows and columns. SQL is used to query and manipulate structured data.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Key Takeaways:

- Database migration tools manage schema changes and data transformations.
- Relational databases organize data into tables with rows and columns.
- Database indexing improves query performance by creating data structures that allow faster data retrieval.